The Evolution of Deep Learning Architectures for Landslide Detection: A Focused Review From CNNs to Hybrid Transformer Networks

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Abstract. The accurate and timely detection of landslides based on remote sensing (RS) imagery is crucial for hazard mitigation, rescue operations and susceptibility mapping. Recent studies have identified three main approaches to image classification: machine learning (ML), object-based image analysis (OBIA) and deep learning (DL). The review explores twenty recent studies on RS-based landslide detection using advanced Convolutional Neural Networks (CNNs) and Transformer architectures. The analysis categorizes the approaches into distinct methodological frameworks such as hybrid data utilization (e.g., optical image and Digital Elevation Model integration), network architecture enhancement (e.g., Swin-UNet variants, CTHNet for hybrid CNN-transformer model applications) and innovative learning methodologies (e.g., knowledge distillation, multi-label classification, transfer learning).We look into how model architectures (e.g., attention-based modules like CBAM, local information aggregation) and tailored loss functions tackle issues such as data imbalance, cross-regional prediction and the precise mapping of landslide boundaries. We found a clear tendency that the models leveraging both local and global feature extraction techniques perform better in various geographical and geological environments. It also stresses the necessity of developing large-scale standardized multi-modal datasets and improved model interpretability for operational use.

Keywords: Landslide Detection; CNN; Vision Transformers; Hybrid Models; Fusion Models

1 Introduction

Landslides are ubiquitous and constitute the most destructive category of natural hazards, necessitating precise and swift delineation of boundaries to facilitate targeted emergency responses. Due to global change (climate and seismic activity intensification, increased human exposure associated with population growth in

mountainous areas, urbanisation, etc.), landslides occur more frequently and cause increasingly higher economic damage as well as many casualties. Accurate determination of the spatial extent and precise boundaries of these natural occurrences is a high priority for decision-makers, post-disaster humanitarian aid programs and long-term land-use planning. This necessity for timely and accurate information has propelled disaster science from traditional manual methodologies, which are largely qualitative, to automated techniques that yield quantitative data. The current study aims to investigate the trends and prospects of automated landslide detection methods based on remote sensing (RS) and deep learning (DL) approaches for landslide inventory mapping.

1.2 Remote Sensing

Remote sensing (RS) technologies are the backbone of landslide detection globally. They enable macro-scale and objective data collection for hazard management, which was hard to implement using inefficient and high-risk field surveys. The current generation of RS platforms provides abundant and diverse data sources. These sources can be categorized into two broad types: (1) Optical/multispectral data which include very high-resolution (VHR) optical images from UAVs and/or satellites (e.g., Planet, Google Earth, GF-6) and multi-spectral images such as Sentinel-2 or Landsat; and (2) Topographic data such as Digital Elevation Models (DEMs) or slope layers. The landslide signatures include the spectral (soil/rock exposure), textural (debris dispersion) and topographic aspects. However, to leverage this data, models need to address several fundamental issues: the ability to combine multi-modal sources (e.g., RGB optical bands and non-spectral DEM data) for enhanced feature representation; poor spatial and spectral contrast between landslide regions and the vegetated background; severe class imbalance between scarce landslide pixels (foreground) and abundant background clutter. Accordingly, the transition from the RS multi-channel inputs to precise landslide pixel discrimination (termed semantic segmentation) is the principal DL model workflow discussed in this review.

1.3 Landslide Detection Techniques

The first attempts to automate landslide detection from satellite images relied on traditional pixel-based or feature-engineering approaches. These methods typically employed spectral thresholds, simple band ratios, or classic machine learning (ML) algorithms like Support Vector Machines (SVM) and Random Forests (RF). However, the pixel-based techniques often suffer from the “salt and pepper effect” and do not fully incorporate spatial context. A major advancement was object-based image analysis (OBIA), which segments images into meaningful objects before classification, allowing researchers to incorporate geometric characteristics such as size and shape in addition to spectral properties. OBIA is generally regarded as the first step toward a knowledge-based image analysis approach. However, both methods (ML and OBIA) necessitated significant domain expertise for manual feature extraction and subjective rule-set design, which hampered their adaptability and scalability in different regions or geological settings. Deep learning’s triumph in CV (computer vision) provided a definitive answer: DL networks could learn features hierarchically and abstractly from

raw image pixels, obviating the need for manual feature extraction. Which placed DL-based CNN (Convolutional Neural Networks) architectures as the current state-of-the-art paradigm for performing accurate, reliable and highly scalable landslide semantic image segmentation.

2 Review Methodology

To ensure the scientific quality of the study, as well as focus on the analysis of recent trends in architecture, the following steps were followed. Primary searches were conducted using Scopus, IEEE Xplore and Web of Science databases. Key search strings for literature identification included: "Landslide Detection," "Deep Learning," "CNN," "Transformer," "Hybrid Model," and Remote Sensing. The selected studies had to fulfill the following criteria: (a) Employ automated landslide detection techniques based on remote sensing imagery; (b) Utilize deep learning models for landslide segmentation (CNNs, Vision Transformer and Hybrid architectures); (c) Report sufficient quantitative data for extracting F1-Score and mean Intersection over Union (mIoU). Papers were excluded if they only focused on susceptibility mapping, solely applied existing machine learning methods, or were review papers. Initial searches produced over 100 results, but only approximately twenty studies were retained after abstract and full-text screening. The selection focused on the most recent and methodologically innovative studies published within the last four years. This selection strategy further narrowed the study's scope by focusing on the transition from CNNs to today's state-of-the-art Hybrid/Fusion models.

3 Literature Review

3.1 CNN Based Models

The early and foundational phase of deep learning approaches for landslide detection, utilizing CNNs for pixel-level classification and instance segmentation. Core contributions involve applying established models, such as U-Net [1] and its variants, sometimes in conjunction with traditional machine learning techniques (e.g., SVM or Random Forest) to show basic segmentation capabilities when sample sizes are small. Mask R-CNN [2, 3] enabled the implementation of high-precision transfer learning techniques for the instance segmentation task. The improved Faster R-CNN model's backbone (DarkNet53 [4]) for CNN core implementation focused on stability and speed, solving the vanishing gradient problem in ultra-deep networks. However, the CNN's embedded nature prevents it from obtaining global information when it comes to feature extraction. And forces the research community to look for complementary non-convolutional architectures because of the tendency to confuse landslides with geologically similar backgrounds. A summary of CNN based models for landslide detection is given in table 1.

Table 1. Summary of CNN based models used for landslide detection

Ref No.	Core Architecture	Primary Innovation	Key Metrics	Limitations
[2]	Mask R-CNN (ResNet-50), U-Net	Systematic comparison of instance vs. semantic segmentation models across regions via transfer learning.	Mask R-CNN superior (Dice: 0.71)	U-Net tends to "clump" landslides in proximity.
[3]	Mask R-CNN (ResNet-101)	Exploits transfer learning on small datasets of UAV-acquired digital photographs.	97% (F1-Score)	Relies heavily on high-quality annotation for small sample sizes.
[4]	Faster R-CNN (DarkNet53)	Replaced heavier backbones with DarkNet53 to stabilize training and minimize the vanishing gradient problem.	92.05% (AP), 0.80 (F1-Score)	Accuracy is lower than semantic segmentation models.
[1]	U-Net	Benchmark testing of U-Net against classic ML algorithms (SVM, RF) on small Himalayan satellite datasets.	71.12% (F1-Score)	Performance gains over ML are minimal with small datasets.
[5]	DenseNet	Optimization of DenseNet structure for efficiency and feature density when augmented with 12 environmental factors.	95.05% (F1-Score)	Demonstrated challenge of noise when merging numerous geological factors.

3.2 Vision Transformer (ViT) Based Models

ViT was the next obvious step because CNNs cannot model long-range dependencies due to their inherent design. The research primarily focuses on better harnessing the power of the Transformer model for remote sensing applications. The realization of this possibility during the development of SwinLS, [6] which achieved high accuracy by adapting the hierarchical structure of Swin Transformer to the task. The key debate arose here which data to use: SwinLS showed that sometimes it is better to utilize only RGB bands, because excessive multi-spectral noise dissipates self-attention focus. At the same time, several studies highlighted that ViT is computationally expensive and needs large amounts of training data. The design of the Distilled Swin-Transformer (DST) [7] empowered implementation in resource-scarce environments without much loss of performance, which was facilitated by knowledge distillation. A summary of vision transformer models for landslide detection is highlighted in table 2.

Table 2. Summary of Vision Transformer (ViT) based models for landslide detection

Ref No.	Core Architecture	Primary Innovation	Key Metrics	Limitations
[6]	Swin – Transformer	Optimization of spectral input (uses RGB only) and a two-stage balanced training strategy for class imbalance.	73.99% (F1-Score)	Performance degraded when irrelevant multi-spectral bands were added.
[7]	Distilled Swin-Transformer	Implements Knowledge Distillation to create a lightweight "student" model for improved computational efficiency (lowest FLOPs).	98.16% (F1-Score)	Generalizability to other landslide types (e.g., earthquake) requires further study.

3.3 Hybrid/ Fusion Models

The vast majority of state-of-the-art solutions fall into the Hybrid/Fusion category, clearly indicating that no single model type is sufficient for landslide detection at a high level. They synergistically utilize the local feature learning ability of CNNs and the global context modeling strength of Transformers. Key synthetic innovations can be categorized into four areas: (1) Architectural blending, such as embedding a lightweight Transformer (CTHNet) [8] into a CNN encoder-decoder to jointly exploit multi-scale features; (2) Targeted feature enhancement, using specifically designed modules (LIAM/MFFLCM [9]or CASI/SDE [10]) to refine local details and preserve boundaries; (3) Data modality fusion, employing separate branches without shared weights in the backbone network (FFS-Net [11]) to extract semantic information from HRSI and DEM inputs; and (4) Knowledge integration, combining deep feature outputs with an OBIA ruleset for refinement [12, 13]. A summary of hybrid or fusion models for landslide detection is highlighted in table 3.

Table 3. Summary of Hybrid/Fusion models for landslide detection

Ref No.	Core Architecture	Primary Innovation	Key Metrics	Highlights
[14]	BiSeNet + DenseNet	Lightweight Architecture: Replaces BiSeNet's Context Path with DenseNet blocks for parameter efficiency while retaining feature density.	90.06% (F1-Score)	Achieves high accuracy with the fewest parameters and lowest memory consumption.
[12]	ResU-Net + OBIA	Fuses CNN probability map with external Rule-Based OBIA rulesets (using indices) for post-processing refinement of landslide boundaries.	72.49% (mIoU)	Demonstrated 22% mIoU improvement over ResU-Net alone via knowledge integration.

[10]	SwinUNet	Enhanced SwinUNet with CASI (Channel/Spatial Attention) and SDE (Spatial Detail Enhancement) modules to fix boundary loss in Transformer decoder.	83.45% (mIoU)	Achieves superior boundary accuracy (lowest Hausdorff Distance) over competing U-Nets/TransUNets.
[15]	Swin + CLIP	Dual-Coded Network: Integrates "Large Model Prior Knowledge" from a pre-trained CLIP-based encoder via a PKI module for semantic enhancement.	0.882 (F1-Score)	Leverages visual-semantic understanding from general computer vision large models.
[11]	Siamese CNN (DeepLabV3+ Backbone)	Heterogeneous Data Fusion: Uses a Siamese network to separately and intelligently extract/fuse features from HRSI and DEM data for blurred old landslide detection.	0.67 (mIoU)	Designed specifically to improve detection accuracy for challenging visually blurred old landslides.
[13]	Swin-Transformer + BiLSTM	Novel application of Multilabel Classification (MLC) using BiLSTM to model inter-object dependencies (e.g., between landslide and bare soil labels).	97.56% (F1-Score)	Captures spatial relationships beyond binary classification tasks.
[8]	CNN + Transformer	CNN-Transformer Hybrid Encoder (VGG variant + MLT module) designed for joint local and global feature extraction in loess plateau regions.	72.81% (F1-Score)	Explicitly designed to handle the complex, fragmented terrain features of Loess Plateau regions.
[16]	Swin-Transformer	Fusion of multi-source data (RGB+DEM) optimized with a Boundary Loss Function (BLF) to improve boundary delineation in unbalanced data.	75.6% (F1-Score)	Highlights the importance of boundary-aware loss functions for segmentation outputs.
[17]	U-Net	Feature Enhancement Framework combining an Attention module and a Multiscale module within the convolutional U-Net structure.	0.797 (mIoU)	Demonstrates the continued value of optimising traditional CNN U-Net structures with refined attention blocks.
[18]	Swin-T, EffNetV2,	Ensemble Strategy combining outputs of	80.41% (F1-Score)	Represents the peak performance

	SegFormer	multiple leading models with advanced training techniques (separated normalization, progressive self-training, Lovasz loss).		strategy in a large-scale international benchmark competition.
[19]	CNN/Cascade/ Fusion	Benchmarks different semantic segmentation frameworks (Fully Convolutional, Cascade, RDFNet) for robust landslide detection.	RDFNet: 0.67 (F1-Score)	RDFNet showed superior accuracy by fusing both RGM and Depth (DEM) features.
[9]	Swin-T + Custom Modules	Designed custom MFFLCM and LIAM modules to explicitly force multi-scale feature fusion and local information aggregation.	90.7% (F1-Score)	Explicitly targets better feature fusion for small objects and fine-grained details.
[20]	ResU-Net + Transformer	Integrates Transformer encoder into the ResU-Net backbone to model global context, using CBAM for attention-based feature refinement.	87.73% (F1-Score)	Proved feasibility of combining CNN and Transformer on small sample sizes with attention.

4 Limitations and Future Direction

The major limitation is poor general generalization ability. Many models are domain-sensitive and cannot be applied across different types of events (e.g., rainfall-induced versus earthquake-induced landslides). Targeting blurred or old landslides requires different feature sets compared to the newly emerged ones. Constructing large-scale multi-modal datasets and improving cross-domain transfer learning schemes would be necessary to address these issues. However, the more sophisticated or complex the model is, the lower the accuracy of the boundary features and other comprehensive aspects. Future studies should focus on developing lightweight models through distillation and pruning techniques. A major gap in the application of remote sensing techniques is the lack of integration of synthetic aperture radar (SAR) data, which is crucial because it allows all-weather and day-night observations. SAR data is a prerequisite for InSAR and pre-collection of relevant datasets is imperative to ensure operational robustness. Finally, the review calls attention to the need for better model explainability (XAI). Since complex hybrid models function as black-box systems, decision-makers need to understand and trust the outputs to deploy them in operations. The next research frontier must focus on bridging this gap with explicit knowledge (e.g., OBIA) and semantic-level understanding (e.g., geological graph models).

5 Conclusion

The systematic literature review established an evolutionary trend from conventional Convolutional Neural Networks (CNN) methods to Deep Learning (DL) fusion approaches in the automated detection of landslides. The review regards the

Hybrid/Fusion paradigm as crucial for successful feature extraction because it combines the local geometric precision of CNNs with the global contextual awareness that transformers provide. Current studies harness the synergy of HRSI and DEM data alongside multiple feature extraction techniques to achieve optimal segmentation metrics across various complex scenes (e.g., AST-UNet, FFS-Net).However, the high computational complexity and poor cross-domain generalization of models remain key challenges for landslide detection implementations. Therefore, the next frontier in this research field should focus on developing lightweight models through distillation and pruning techniques and emphasize domain adaptation and explicitly integrating geometric prior knowledge to obtain accurate real-time globally applicable landslide inventories.

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